

DocUCare

Project Team

Wasif Mehboob 21I-1766
Abdul Moiz 21I-2687
Ahsan Waseem 21I-2696

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Supervised by

Mr. Saad Salman

Co-Supervised by

Dr. Hasan Mujtaba



Department of Data Science

**National University of Computer and Emerging Sciences
Islamabad, Pakistan**

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Contents

1	Introduction	1
1.1	Problem Statement	1
1.2	Scope	2
1.3	Modules	2
1.3.1	Module 1	2
1.3.2	Module 2	3
1.3.3	Module 3	3
1.3.4	Module 4	3
1.4	User Classes and Characteristics	4
2	Project Requirements	5
2.1	Use-case/Event Response Table/Storyboarding	5
2.1.1	Use Case 1: Initial Data Collection by Chatbot	5
2.1.2	Use Case 2: Online consultation	6
2.1.3	Use Case 3: Schedule an On-Site Consultation	7
2.1.4	Use Case 4: Transcription Generation and Review	8
2.2	Functional Requirements	9
2.2.1	Module 1	11
2.2.2	Module 2	11
2.2.3	Module 3	11
2.2.4	Module 4	11
2.3	Non-Functional Requirements	11
2.3.1	Reliability	11
2.3.2	Usability	12
2.3.3	Performance	12
2.3.4	Security	12
2.4	Domain Model	14
3	System Overview	15
3.0.1	Functionality:	15
3.0.2	Context:	16
3.0.3	Design:	16
3.0.4	Background Information:	17

3.1	Architectural Design	17
3.1.1	Description:	18
3.2	Design Models	18
3.2.1	Extended Use Cases:	18
3.2.2	Activity Diagram:	21
3.2.3	Sequence Diagram:	29
3.2.4	Class Diagram:	35
3.3	Data Design	36
3.3.1	ERD:	36
3.3.2	Description:	36
4	Appendix	41
4.1	Multilingual (English and Urdu).	41
4.2	Integration with Electronic Health Records (EHR) systems.	41
	References	43

List of Figures

2.1	UC 1	5
2.2	UC 2	6
2.3	UC 3	7
2.4	UC 4	8
2.5	Domain Model	14
3.1	MVC Model	17
3.2	UC 1: AD 1	21
3.3	UC 2: AD 2	22
3.4	UC 3: AD 3	23
3.5	UC 4: AD 4	24
3.6	UC 5: AD 5	25
3.7	UC 6: AD 6	26
3.8	UC 7: AD 7	27
3.9	UC 8: AD 8	28
3.10	UC 1: SD: 1	29
3.11	UC 2: SD: 2	30
3.12	UC 3: SD: 3	30
3.13	UC 4: SD: 4	31
3.14	UC 5: SD: 5	32
3.15	UC 6: SD: 6	33
3.16	UC 7: SD: 7	34
3.17	UC 8: SD: 8	34
3.18	Class Diagram	35
3.19	ERD	36

List of Tables

Chapter 1

Introduction

As we all know that healthcare industry is rapidly moving forward with the incorporation of artificial intelligence. The purpose of the healthcare sector is to enhance the quality of care rendered to patients as well as to improve the operations of the administration. One such task that can easily be performed using AI is medical transcription and conversational artificial intelligence. This project seeks to come up with a tool that will do more than just produce precise records of doctor patient interactions but will also simulate a voice based assistant for boosting the documentation process and cutting down on human related efforts and inaccuracies. [1] [2] [3]

1.1 Problem Statement

The healthcare sector is constantly fighting with the issue of effectively capturing and documenting doctor and patient encounters. Conventional transcription techniques require a lot of time and have too many drawbacks. Such methods can be a burden for the medical staff who has to make sure that details captured are error free. There are problems such as failure to note every detail in during patient consultation, especially when patients talk about their problems. There are many chances of making a mistake in writing the medical terms. Moreover, there is a loss of efficiency with regards to writing and keeping notes, which makes it very hard not to have some vital information about a patient. This makes the assessment quite long, and the planning of the patient care in the years to come inadequate. Human participation is often needed at the beginning of patient data collection, thus wasting time and increasing opportunities for the occurrence of documentation mistakes. Our solution has been developed to overcome these problems by eliminating the labor intensive components of the transcription and patient's initial reports using AI based chatbot and voice-to-text technology. This will ensure minimal involvement of the human being. The software completely removes the necessity of taking down notes man-

ually, increases the precision of the data gathered and minimizes the paperwork required from the physicians so that they can concentrate on giving care to their patients.

1.2 Scope

The scope of the project entails creating an AI system that enables accurate and efficient medical transcription generation through communication between a doctor and a patient either online or onsite. The main functionalities encompasses a speech to text system that captures and writes down all information that was captured in the medical dialogue and regardless of some challenges like background noise, medical words, etc. This solution will also incorporate speaker split to determine the talker as the doctor or the patient during recording for clear transcription later. Also an audio based chatbot will be added to assist in patient friendly session of collecting initial information or data from patients prior to the doctor attending to them. This decrease the necessity for doctors or assistants to ask questions manually and facilitates the completion of the consultation. The system will be integrated with telemedicine platform for online consultation. This will be a very convenient solution for patients that cannot travel or for those who live in rural areas where professional or relevant doctors are not available. For those patients who want to have consultation with the relevant doctors onsite, the system will also be integrated with Google Maps so that they can find the relevant doctors nearby for onsite consultation. The initial reports will be shared with the doctors for analysis before consultation. The system will incorporate speech recognition technology and natural language processing (NLP) to generate initial and accurate reports of patients through their interaction with the chatbot. The reports will then be shared with the relevant doctor for further analysis. During patient's and doctor's conversation, transcription will be generated. The errors will be rectified in both the initial reports and transcripts to avoid any misconception or discrepancies. The system will be designed and available as a service to patients and healthcare providers.

1.3 Modules

1.3.1 Module 1

Module 1 is the development of chatbot which will be integrated with our system and UI/UX. The patient will first interact with it and share their relevant data by answering the questions asked by our bot. (Client Web/Mobile App)

1. Patient login/registration and interface.

2. Integration of chatbot.
3. Interaction with the chatbot.
4. Patient's initial data collection.
5. Accurate reports generation.

1.3.2 Module 2

Patient will decide whether to have consultation with relevant doctor online or onsite. (Client Web/Mobile App)

1. Selection or recommendation of relevant doctor.
2. Initial reports sharing with the doctor.
3. Telemedicine platform integration for online meeting.
4. Google Maps integration for searching relevant doctor for onsite meeting.

1.3.3 Module 3

Doctors registered on our system will have different interface where they can see and analyze patient's initial reports and appointment request. (Admin Web/Mobile App)

1. Doctor login/registration and interface.
2. Patient's initial reports analysis.
3. Meeting schedule (online or onsite).

1.3.4 Module 4

During the conversation of patient and doctor, either online or onsite, transcription will be generated accordingly. (Admin and Client Web/Mobile App)

1. Speech to text conversion.
2. Speaker diarization.
3. Filtering and rectifying noise and errors.
4. Accurate transcript generation for both the patient and doctor.

1.4 User Classes and Characteristics

User class	Description
Patients	Patients will serve as the primary users who will require medical consultations. They will interact with the system for online and onsite consultations with doctors. The integrated chatbot will be used for initial data collection. The system will then help them find relevant doctors or schedule consultations via telemedicine platforms, or even locates nearby healthcare providers via Google Maps.
Doctors	Doctors will use the system for online or onsite consultations and rely on the transcription generated during their conversations with patients. They will also review preliminary reports prepared by the chatbot and verify or amend them as necessary.
Admins	Admins would manage and take care of the system to function the system effectively, securely, and according to the defined policies. They will play an important role of maintaining the integrity of the system and would support healthcare professionals and patients.
System Developers and IT Support	The systems developers and the IT support team will make sure that the system's platform is up and safe. To that end, the IT support team shall troubleshoot, update the system, and integrate third-party tools such as telemedicine platforms and Google Maps.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case/Event Response Table/Storyboarding

Actors:

- Patient: seeks consultation with the chatbot, views report, and submits an online or onsite consultation request to relevant doctor.
- Doctor: attends to the patient's details, medical comment, gives consultancy, and accesses or modifies the transcription generated.
- Telemedicine Platforms: Inclusions of Online Consultations.
- Google Maps: Searching doctor nearby for on-site consultation.
- Audio-based chatbot: Patient interacts with chatbot for initial reports generation before consultation.

2.1.1 Use Case 1: Initial Data Collection by Chatbot

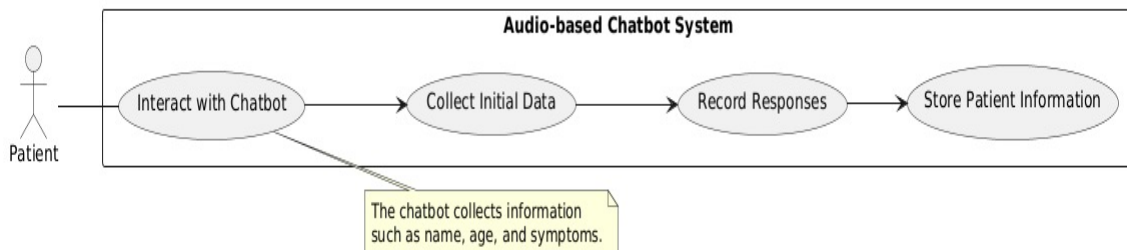


Figure 2.1: UC 1

- Actor: Patient
- Objectives: Audio-based chatbot for preliminary information gathering from the patient before consultancy.
- Pre-Condition: Patient logged into the system.
- Steps:
 1. Patients discuss with the chatbot.
 2. Chatbot questions the patient using a set of questions (name, age, symptoms, medical history, diet plan, routine, etc.).
 3. The patient will talk, and the system will record responses accordingly.
 4. The information is consequently stored in the system and later analyzed by the doctor.
- Post-condition: The patient's initial data is stored and retrievable for the doctor to view before or during consultation.
- Exception: If the chatbot couldn't understand the patient, it asks for clarification.

2.1.2 Use Case 2: Online consultation

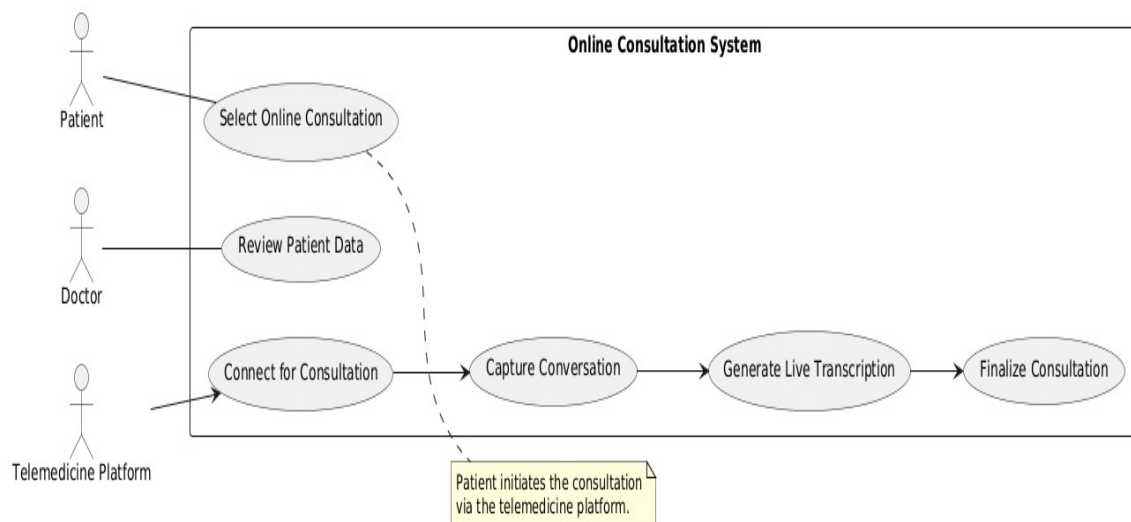


Figure 2.2: UC 2

- Actor: Patient, Doctor
- System Actor: Telemedicine Platform

- Objectives: To prepare for an e-consultation between the patient and the physician.
- Pre-condition: Both the patient and doctor should be logged in to the system using an integrated telemedicine platform.
- Steps:
 1. The patient chose the online consultation and logged onto the telemedicine service.
 2. Doctor reviews the patient's preliminary data.
 3. The system records the conversation and transcribes the recording.
 4. Doctor and patient discuss the case.
 5. Doctor then gives opinions or prescriptions.
 6. The transcription is processed and diarized, separating the contributions of doctors and patients.
- Post-condition: The consultancy session has been closed and the transcription report generated afterwards. Transcription shared with the doctor and patient and saved for probable future reference.
- Exception: The system will try to reconnect, or reschedule meeting if the attempt fails because of connectivity issues.

2.1.3 Use Case 3: Schedule an On-Site Consultation

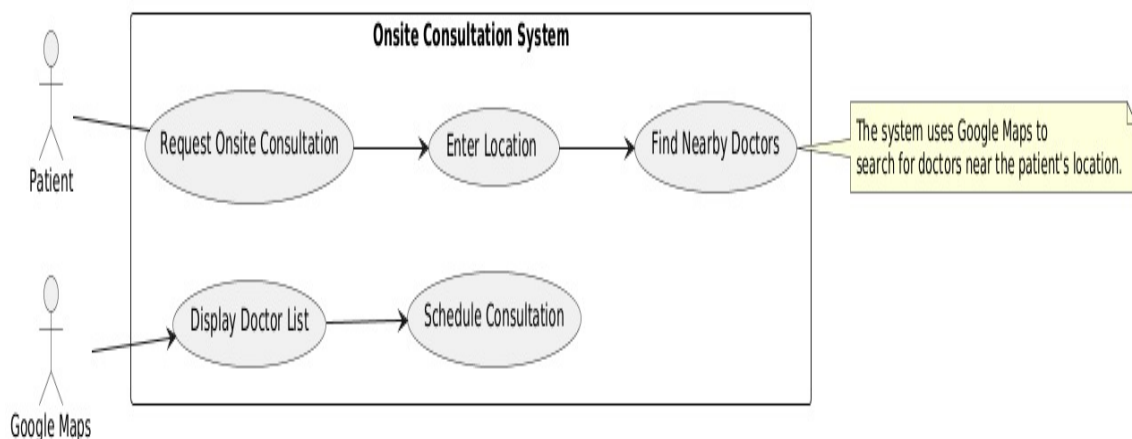


Figure 2.3: UC 3

- Actor: Patient
- System Actor: Google Maps

- **Objective:** Enable the patient to find doctors around them who can be consulted onsite, hence leveraging Google Maps.
- **Precondition:** The patient attempts to access the system and requests an on-site consultation.
- **Steps:**
 1. The patient inputs the information into the system.
 2. The system connects to the Google Maps API/SDK to find nearby doctors.
 3. The system can show the number of doctors available along with the details.
 4. The patient then chooses a doctor and books an on-site appointment.
 5. The system confirms the appointment and forwards related data to the doctor.
- **Aftercare:** Discussion is scheduled on-site, and the physician is notified.
- **Exception:** In this case, it's showing the future online appointment that can be scheduled if no physicians are found locally.

2.1.4 Use Case 4: Transcription Generation and Review

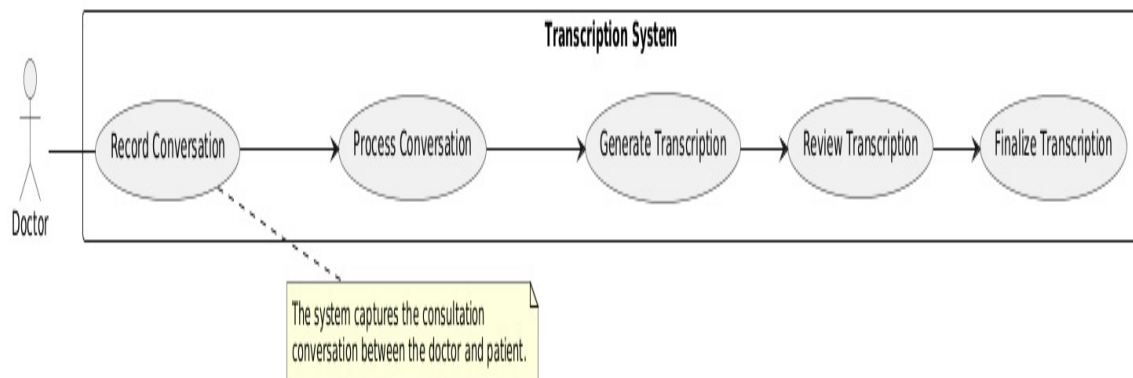


Figure 2.4: UC 4

- **Actor:** Doctor
- **Objective:** Doctor-Patient Conversation Transcription generation presenting the discussion of the doctor and patient's dialogue.
- **Pre-condition:** The doctor has been holding follow-up consultations with the patient.

- **Procedures:**

This system monitors conversations or dialogues between the doctor and patient and prescriptions given by the doctor. The system processes the conversation through speech-to-text technology. The transcription is diarized, separating the doctor's contribution and the patient's input. Transcription is then shared with the doctor and patient. The physician checks the transcription and adds any remarks or other medical remarks. That transcription is then stored within the system's database for future reference.

- **Exception:** If there are still some errors or missing information in transcription, the doctor can manually correct them.

2.2 Functional Requirements

FR 1: Speech-to-Text Transcription

- This would enable the system to transcribe real-life conversations between physicians and patients in printed text with reasonable accuracy.
- The system shall transcribe real-time audio conversations into text.
- The system should remove background noises and other irrelevant sounds during transcription.
- The system should also record the medical terms mentioned in the conversation.
- Every speaking doctor and patient will be associated with their respective conversations when the system is translating.
- It won't err in transcription during pauses and breaks in conversation.

FR 2: Speaker Identification and Diarization

- It would basically help point out who was speaking, if it was a physician or a patient.
- The system must identify speakers by their names or enroll them as the patient or doctor.
- It will hence make an accurate diarization of a multi-speaker conversation.
- There must also be provision for human override in case of misidentification of speakers.

FR 3: Audio-based pre-consultation Chatbot

- The patient will feed preliminary medical information to the chatbot.
- In the system, there should be a chatbot that interacts with patients and retrieves their history and symptoms.
- The chatbot collates the information retrieved and the system makes a pre-sessional report for the doctor to present before the patient.
- This chatbot shall be able to understand and respond in a normal language query.

FR 4: Collaboration with Telemedicine platforms for Online Consultation

- It will be designed to provide practical, online consultative capacity using telemedicine platforms.
- It will thus be combined with telemedicine platforms to offer online consultations.
- The system will keep online consultations for transcription purposes, which the patient agrees to.
- It will record the interactions that occur during the online meetings and maintain a record.

FR 5: Google Maps Integrated into the On-Site Consultation Scheduling Function

- This will enable patients to make appointments for site visits and locate doctors who can treat them.
- The system should allow the search of physicians based on specialty and location.
- It will interact with Google Maps to show nearby doctors who can consult onsite.
- It should also be possible to allow to book doctor-patient appointments on the system.

FR 6: Report Generation and Review

- The main feature shall generate and transfer health reports after the consultation.
- It will give a healthy medical report based on the transcription of the conversation.
- The findings shall be presented and analyzed with the doctor and communicated to the patient.
- The system will deliver error correction so that the transcription becomes a very honest projection of the consultation.

2.2.1 Module 1

Following are the requirements for module 1:

1. FR 3

2.2.2 Module 2

Following are the requirements for module 2:

1. FR 4
2. FR 5

2.2.3 Module 3

Following are the requirements for module 3:

1. FR 6

2.2.4 Module 4

Following are the requirements for module 4:

1. FR 1
2. FR 2

2.3 Non-Functional Requirements

2.3.1 Reliability

- NFR-R1: The system shall have a Mean Time Between Failures of not less than 100-200 hours to minimize downtime during the critical operations that might be those of consultations or data transcriptions.
- NFR-R2: The system shall maintain reliability in generating accurate transcriptions and reports, with minimal downtime.

- NFR-R3: It shall detect errors in key processes, such as transcription and speaker identification automatically, by catching at least 80 to 90 percent of transcription errors.
- NFR-R4: Failure impact should be restricted through the storage of the conversation history up and patient's data to a failure point such that lost interactions are preserved.

2.3.2 Usability

- NFR-U1: The system shall allow a user interface through which a first-time user shall be able to interact with the chatbot and book consultancy through a user friendly interface.
- NFR-U2: The system must integrate seamlessly with telemedicine platforms and Google Maps.
- NFR-U3: The interfaces for both patients and doctors shall be user friendly, ensuring smooth interactions with the chatbot, meeting schedule, and easy report access.

2.3.3 Performance

- NFR-P1: The system shall handle transcription efficiently, even with background noise.
- NFR-P2: The reply by the chatbot should be delivered back to the user in a short time to ensure user-friendly interaction.
- NFR-P3: The system shall ensure high accuracy in speech-to-text transcription, particularly in medical terminology.
- NFR-P4: The system shall be able to handle speaker diarization efficiently for transcription generation.

2.3.4 Security

- NFR-S1: The system shall implement strong user authentication mechanisms, for example multi-factor authentication, to ensure the identity of users when they access the system.
- NFR-S2: The system shall define user roles and permissions which will limit the access to sensitive features and information only to doctors, admin, and patients.

- NFR-S4: The system must log all access and activity, retaining audit logs for 6 months, to ensure traceability in case of security breaches or misuse.

2.4 Domain Model

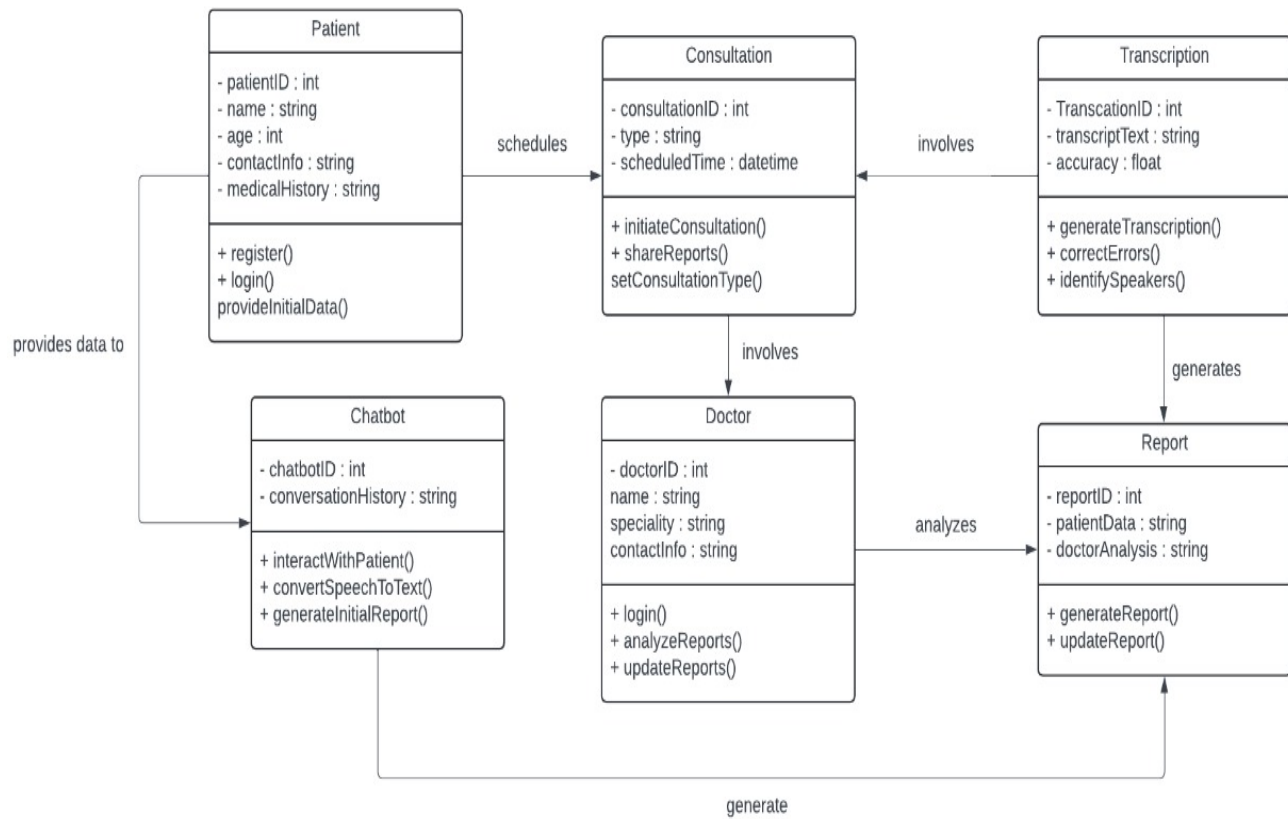


Figure 2.5: Domain Model

Chapter 3

System Overview

It is a patient-doctor interaction system powered by telemedicine capabilities and a chatbot, mostly based on gathering and exchanging patient medical information for consultation and accurate transcription generation from consultation. The system will provide smooth interactions between patients and doctors, collect health information, generate accurate transcription, and enable online discussions through a telemedicine platform or onsite at a physical location.

3.0.1 Functionality:

1. User Authentication:

It allows patients and doctors to log in securely through authentication. It has a feature of registering a new user by filling a form. Different dashboards are received by the patients or doctors on entering their credentials.

2. Chatbot Interaction:

It has a chatbot interacting with the patient to gather some of the initial medical information, such as symptoms, history, diet, and lifestyle information, right from the informal background of the patient. The natural language processing of the chatbot will gather this information accurately with its conversational interface.

3. Report Generation:

The data taken by it from patients would be processed by the system to result in a structured medical report. Such a report indicates the patient's state of health and shall be presented to a doctor for further consultancy.

4. Consultation Options:

Two consultation options for the patients:

Online consultation: It is a telemedicine platform that enables consulting with a doctor using video/audio.

On-site Consultation: Clinic visits to the doctor where the system shall be aligned with Google Maps in locating physicians nearby.

5. **Doctor Interaction:**

Doctor would have the reports of the patient, and they can analyze data collected and schedule a consultation with the patient either online or onsite. They can also view patient history for better analysis.

6. **Speech to Text and Transcription:**

The system captures all the conversations between the doctor and patient during onsite or online consultations. Speech is converted into text, and then the transcript of conversation is generated and shared with both the doctor and the patient and saved for their record keeping and follow up purposes.

3.0.2 **Context:**

It will exist in the health space, trying to improve communications between patients and doctors whose friction around data sharing and communication will be reduced. It will avail of the modern technologies of chatbots, NLP, speech-to-text transcription, and telemedicine platforms in an attempt to reach out to more patients with efficient consultations.

3.0.3 **Design:**

Several interconnected modules make up the system's design:

1. **Easy-to-use interface:** Patients as well as doctors interact through a comfortable and intuitive interface, wherein login registration with dashboards managing reports and consultation with the patient are included.
2. **Backend Processing:** This involves processing algorithms that power the chatbot and report generation feature through the processing of structured medical data from the patient's initial input.
3. **Telemedicine Link:** Connects to integrated telemedicine platform which can offer a virtual consultation and Google Maps that can identify nearby doctors for in-person consultations.
4. **Transcription:** Speech-to-texting technology captures conversation and transcribes their documents for the benefit of both parties to share.

3.0.4 Background Information:

This project reflects the general trend of digital transformation in healthcare as telemedicine and AI-based health care systems have been taken to the center. Telemedicine creates some solutions that become inevitable by allowing a reduction of physical visits to hospitals and faster, more effective treatment of the patients. Second, the chatbot technology addresses the issue of AI-based patient intake, making doctors pay less attention to collecting initial data and more to treatment in a further stage. This solution also responds to the increasing need for telehealth services, especially in instances where onsite visits may not be possible—for example during a pandemic or by patients in quite far areas. This development underscores a necessity to integrate medical expertise with technology when developing new tools that are expected to improve health-process-related factors, upgrade the patient's experience, and make doctors rely on time-to-date data in making accurate diagnoses.

3.1 Architectural Design

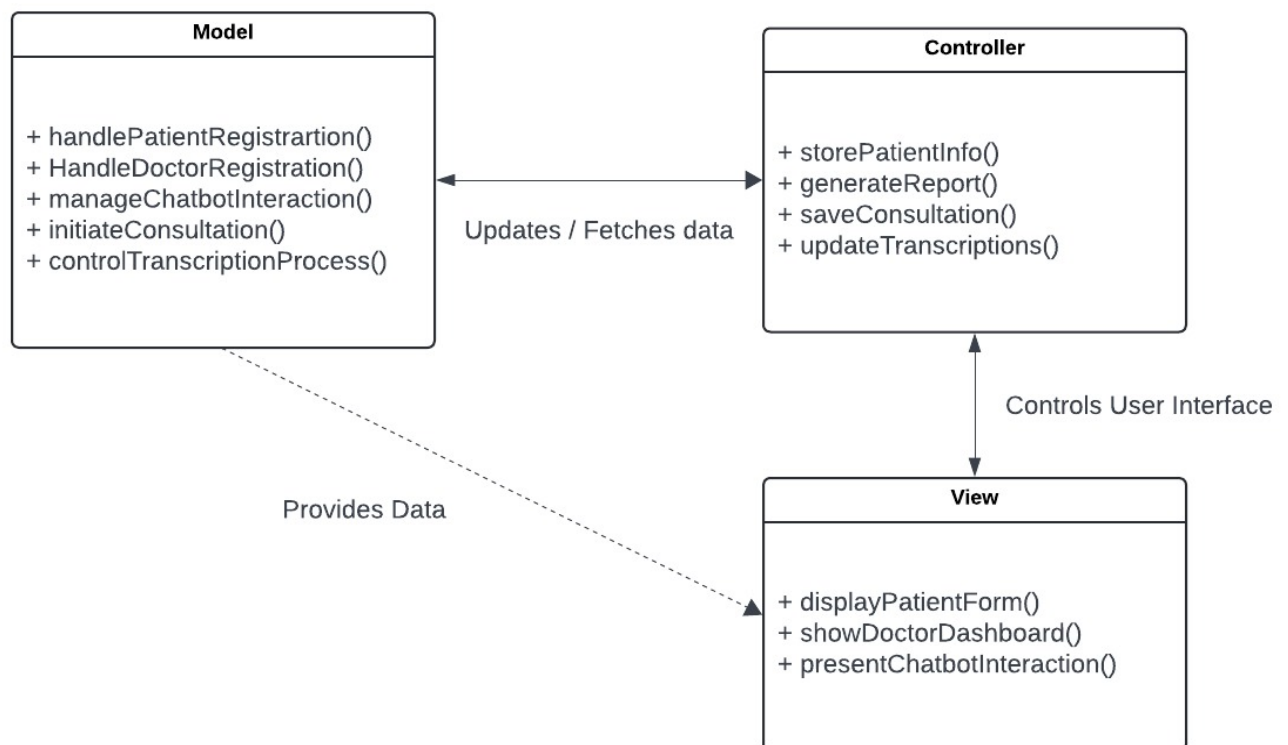


Figure 3.1: MVC Model

3.1.1 Description:

Actually, the MVC architecture is used in this project to make the interaction between the different components involved here just so efficient.

1. Model:

Component The component is in charge of managing the application's data and business logic. It consists of, among other functions including:

HandlePatientRegistration(): Patient registration procedures are under control.

HandleDoctorRegistration(): It takes charge of registering doctors.

ManageChatbotInteraction(): Allow interaction with the chatbot.

InitiateConsultation(): Initiates the consultation process.

ControlTranscriptionProcess(): It controls the transcription of doctor-patient transactions.

2. View:

This relates to the user interface and how data should be graphically presented to the users.

DisplayPatientForm(): It displays the patient registration form.

ShowDoctorDashboard(): Shows the dashboard of the doctor with information concerning the patients and reports.

PresentChatbotInteraction(): Draws the interaction interface for the chatbot.

3. Controller:

The controller acts as an intermediary between the model and the view. This controller receives user inputs and transforms those into consistent with the model.

StorePatientInfo(): Store the pulled patient information.

GenerateReport(): Generates medical reports from data gathered.

Save Consultation(): Saves the consultation details.

UpdateTranscriptions(): Updates transcriptions generated during the interview.

3.2 Design Models

3.2.1 Extended Use Cases:

1. Patient and Doctor Login:

Login Page Appears: With the very opening of an application to a patient or doctor, the second login page appears. It has fields, username and password for the credentials.

Input Credentials: The user enters their login credentials.

Error and Retry on Failed Authentication: If the entered credentials are invalid then an error message is given saying "Invalid username or password" and further entry by the user is allowed to retry. There is also the sign up page where new users may even register if they are not yet found in the system. After clicking on that, users are redirected to the registration page to create account.

Correct Authentication: Once the user has logged in, the system authenticates credentials and takes the relevant person to their dashboard, that is, patient dashboard or doctor dashboard, respectively.

2. Chatbot Interaction:

Activation of Chatbot: After the patient login, the system begins activating a chatbot interaction.

Data gathering: This is the start of the chatbot in welcoming the patient and asking a set of questions to obtain the prime data including the symptoms, dieting habits, medical history, lifestyle, etc.

Interactive Conversation: The chatbot will converse in a natural language, which enables the patient to input data smoothly, since it can understand different variations in the patient's responses.

3. Processing Initial Data and Report Generation:

Data Processing: The chatbot will then process the information about the patient it has gathered and eventually produce a structured initial medical report.

Report Sharing Option to Display: After the report has been generated, a share option will be displayed on the screen through which the patient can share the report with the doctor for analysis.

4. Consultation Options (Online or Onsite):

System will then provide the patient with two options to choose the type of consultation they prefer: it could be online, hence telemedicine, or onsite, meaning a physical visit.

Report Sharing: After a patient has selected a type of consultation, he or she can thereafter share the report with any of the selected doctors for either online or onsite consultancy.

5. Doctor Accesses and Analyzes Report:

Report to Doctor: The report submitted is forwarded by the system to the doctor.

Analysis of Patient Data: The patient's medical history and the symptoms described in the report analyzed by the doctor.

Consultation Scheduling: The doctor will schedule an online or onsite consultation with the patient following the analysis.

6. Online Consultation via Telemedicine Platform:

Telemedicine Platform Connection: A secured connection in the telemedicine platform connects a patient remotely with a doctor, such as a video call or chat, for an online consultation.

Consultation Interaction: The patient and the doctor are talking; the system captures the conversation.

Consultation Closes: When the consultation is complete, the link is closed, and all post-consultation advice is documented.

7. Onsite Consultation with Google Maps Integration:

Doctor Search through Google Maps: The system will likely enable a patient with a need for onsite consultation to seek suitable doctors based on location.

Appointment Scheduling: The patient selects a convenient doctor from the list, books an appointment, and sends the report to the doctor.

Confirmation of Appointment: Confirmation occurs, with ways or reminders of the actual onsite visit with doctor.

8. Consultation Transcript (Online and Onsite):

Audio Recording of Doctor-Patient Conversation: The system records the audio of the consultation session either online or onsite.

Speech to Text Converting: The conversation is automatically made in a text format through speech recognition.

The system will also develop a comprehensive consultation transcript that it will make available to both the doctor and the patient.

3.2.2 Activity Diagram:

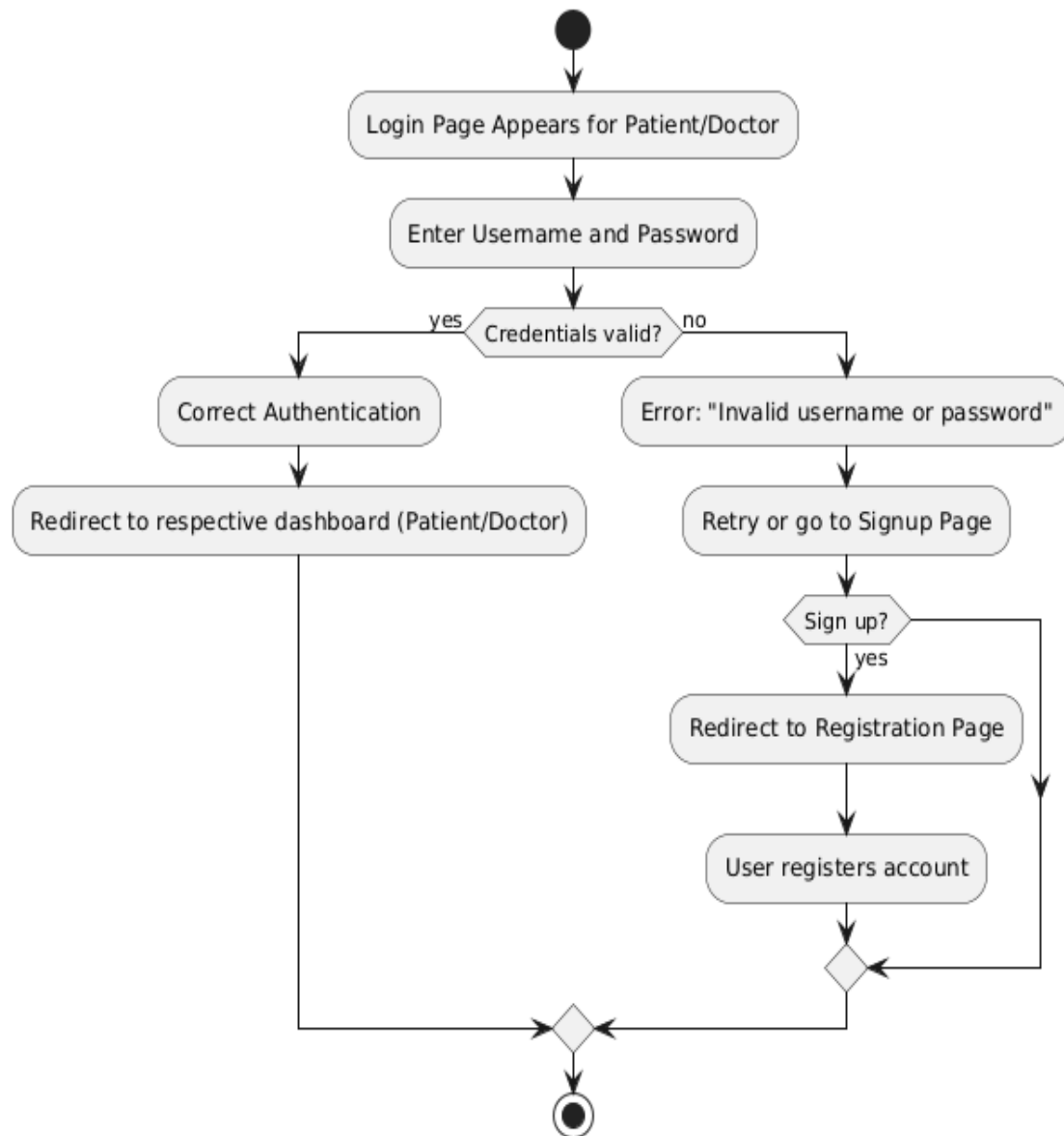


Figure 3.2: UC 1: AD 1



Figure 3.3: UC 2: AD 2

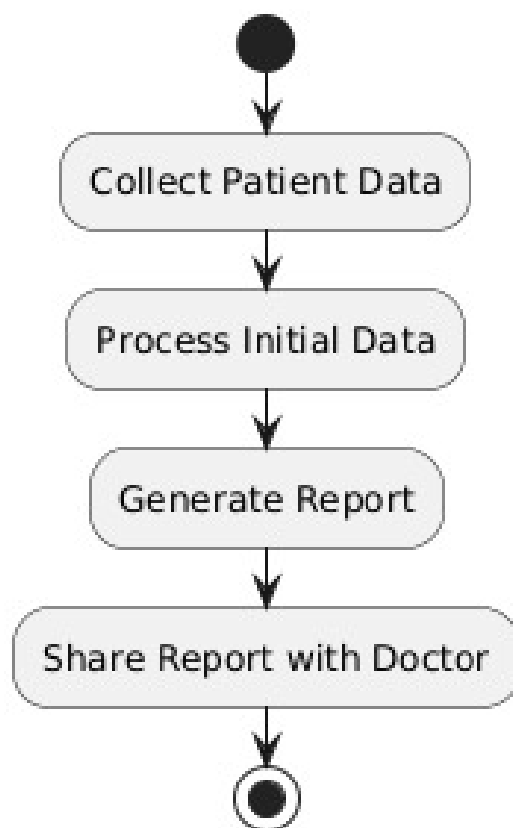


Figure 3.4: UC 3: AD 3

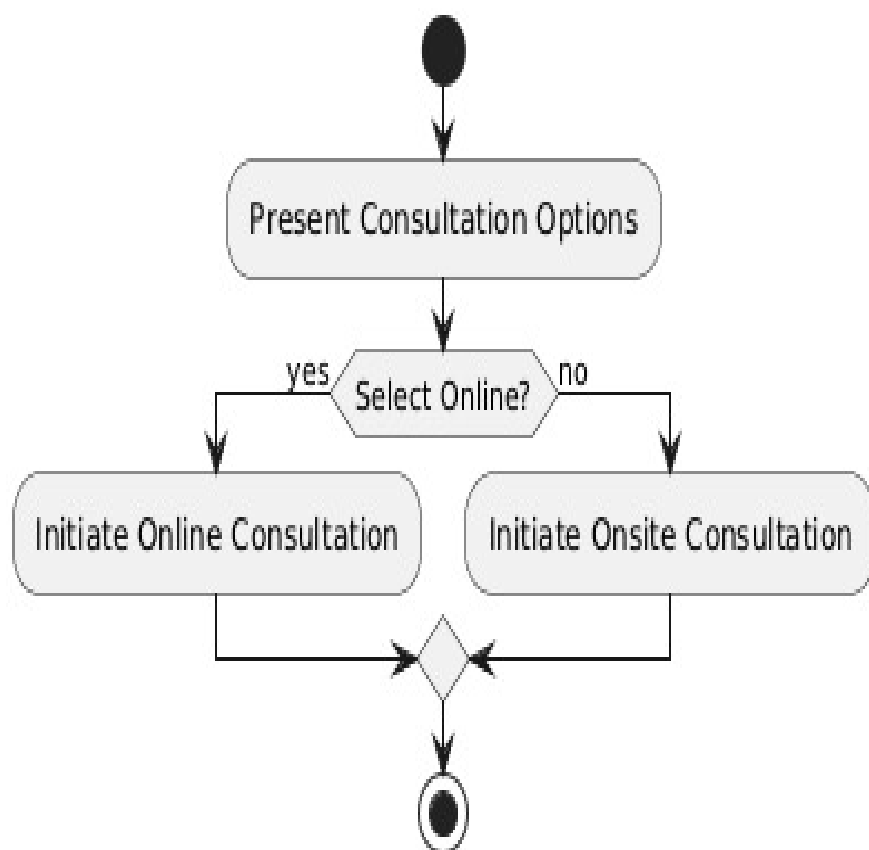


Figure 3.5: UC 4: AD 4

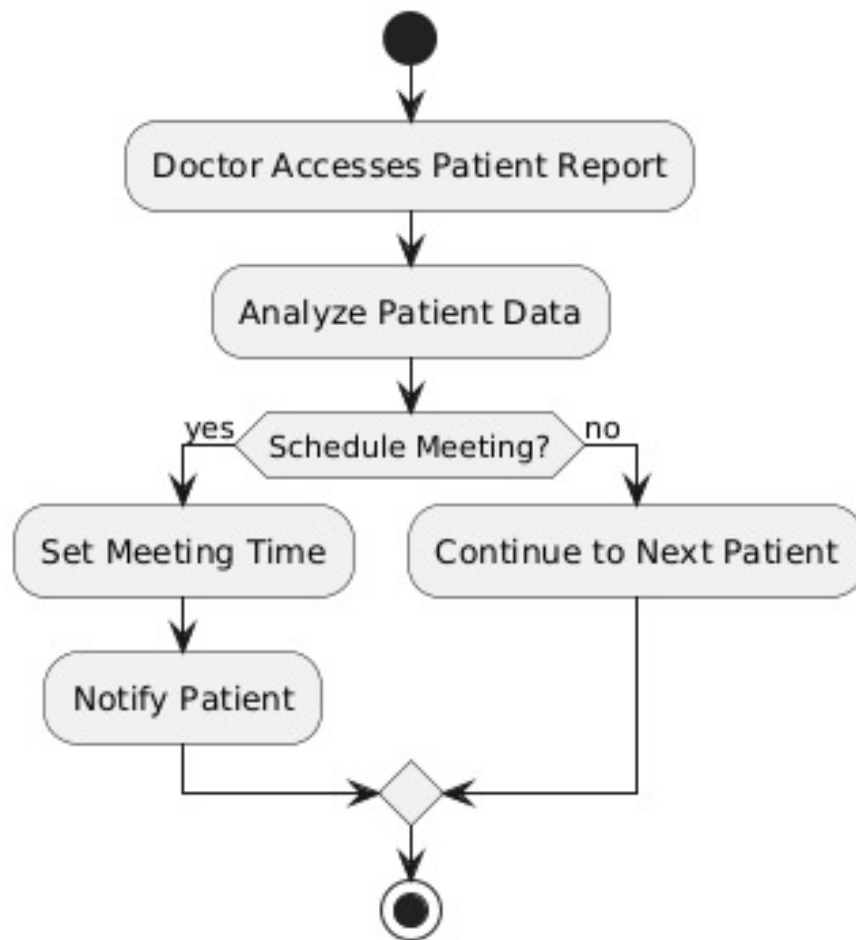


Figure 3.6: UC 5: AD 5

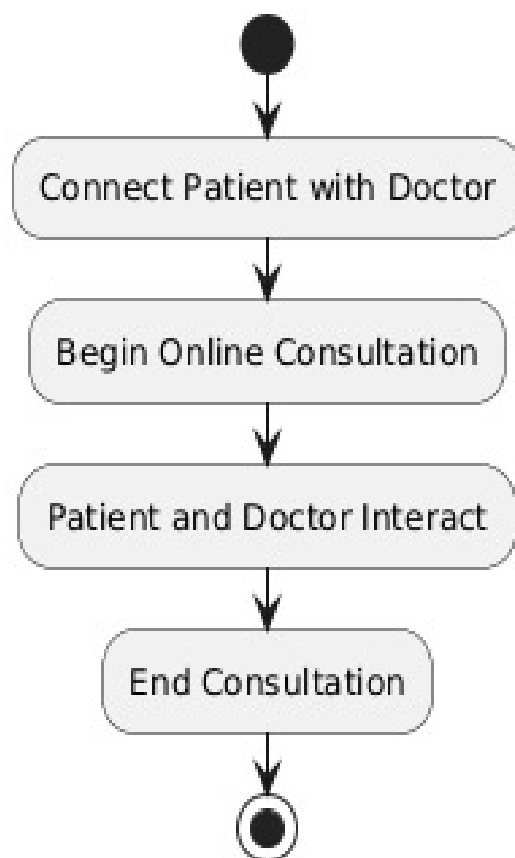


Figure 3.7: UC 6: AD 6

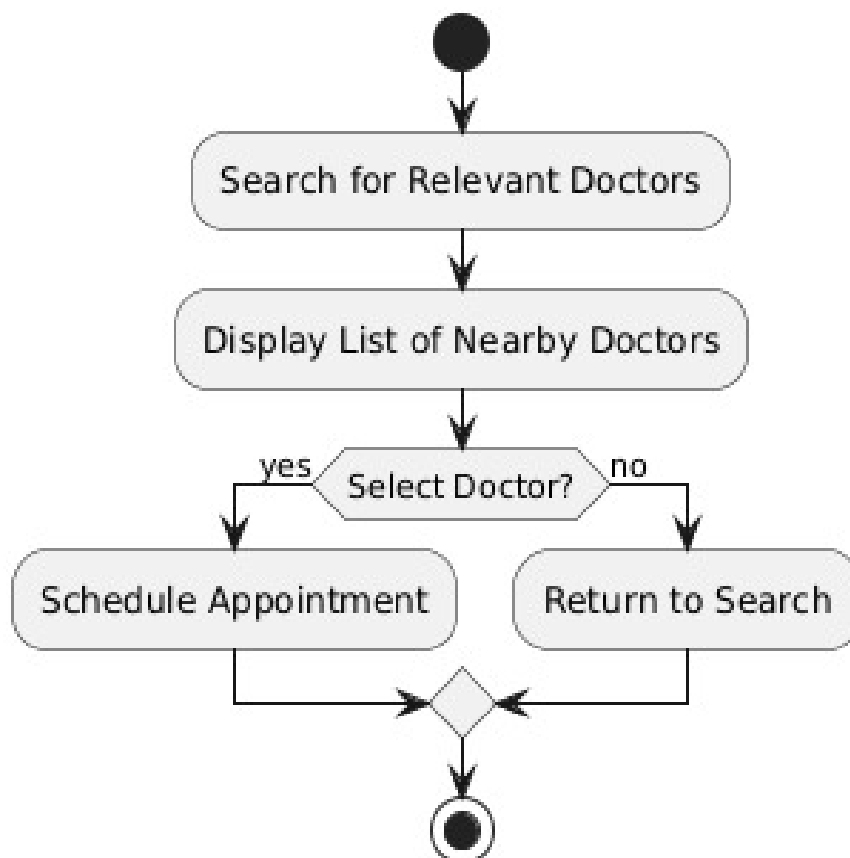


Figure 3.8: UC 7: AD 7

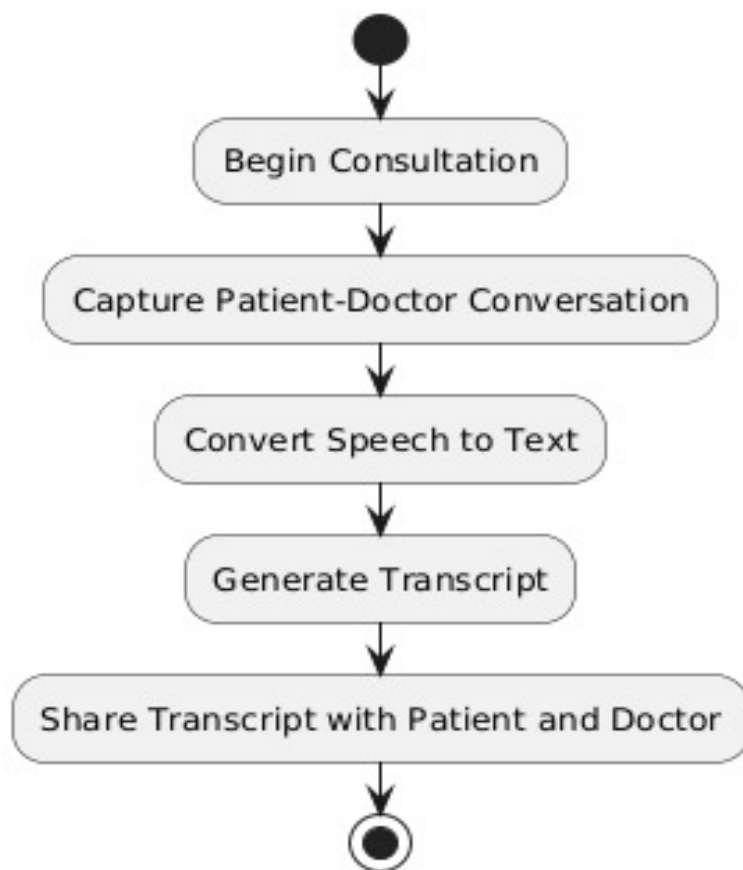


Figure 3.9: UC 8: AD 8

3.2.3 Sequence Diagram:

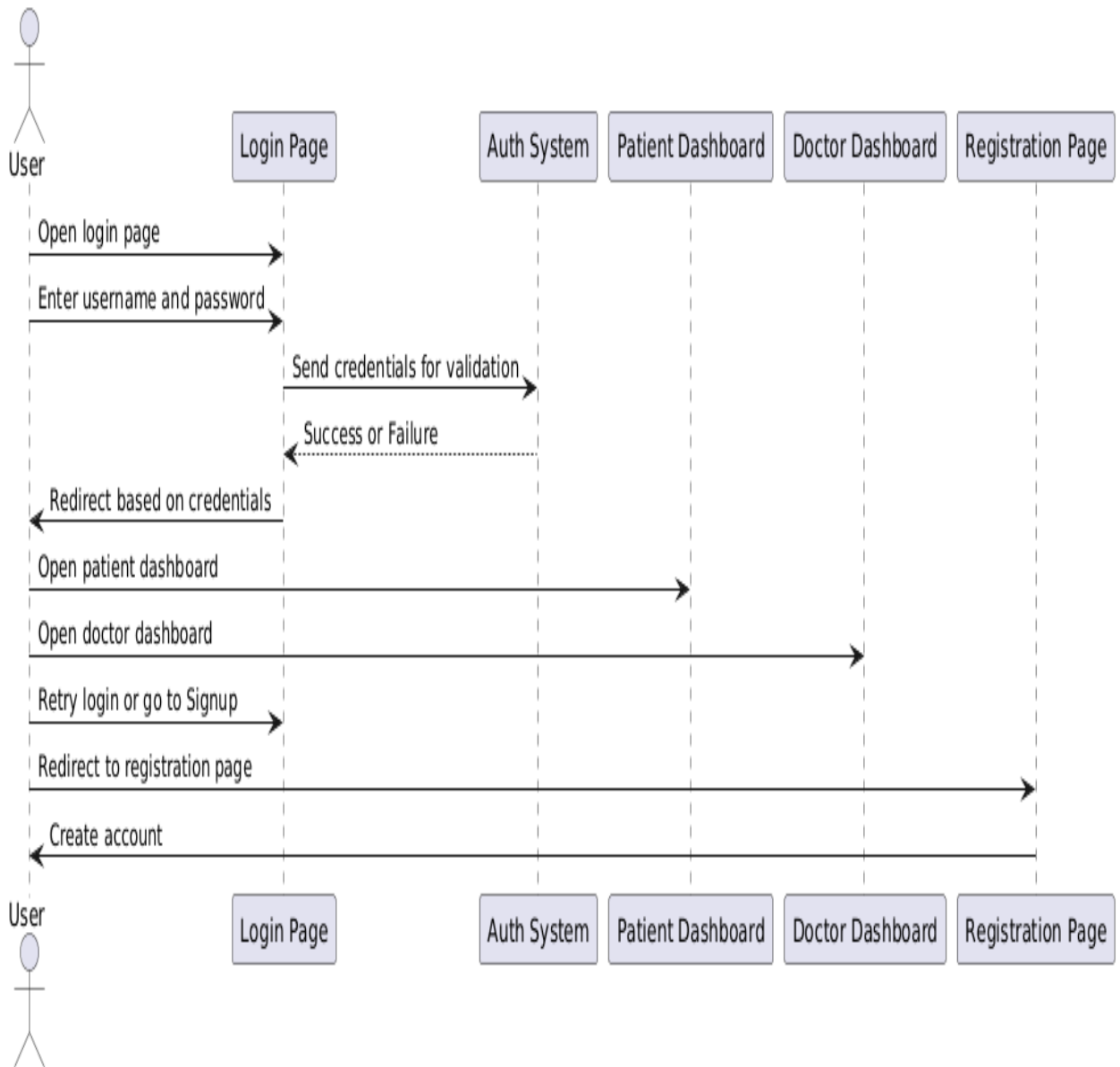


Figure 3.10: UC 1: SD: 1

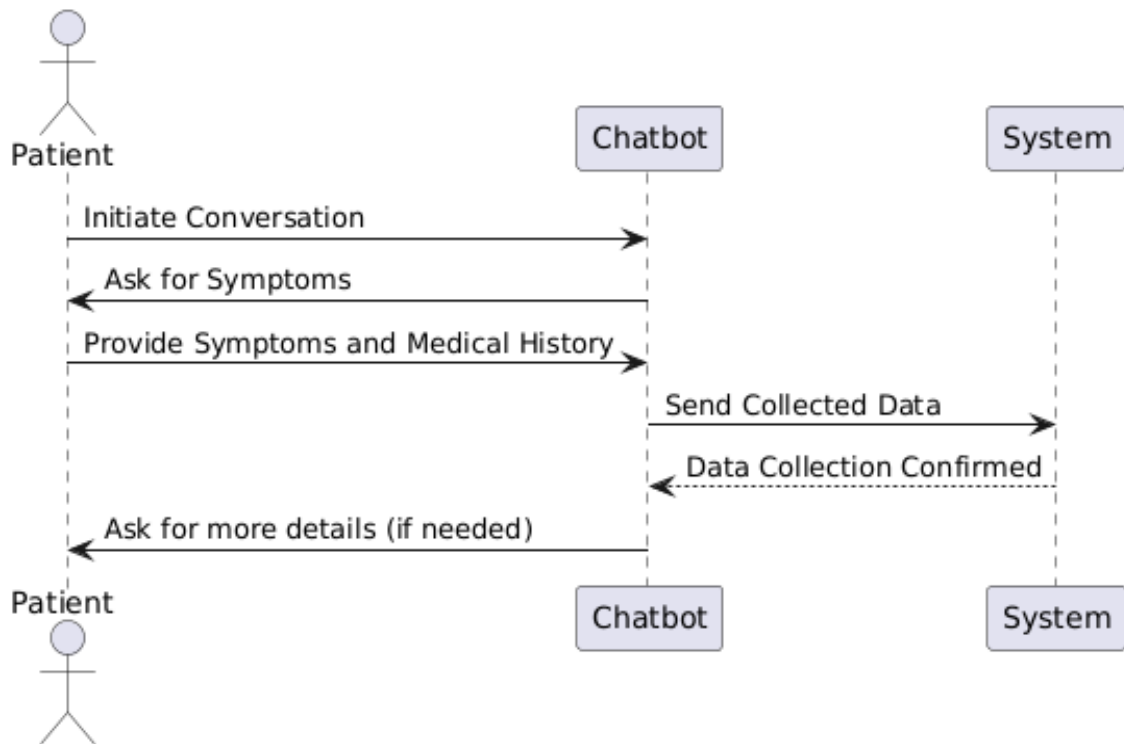


Figure 3.11: UC 2: SD: 2

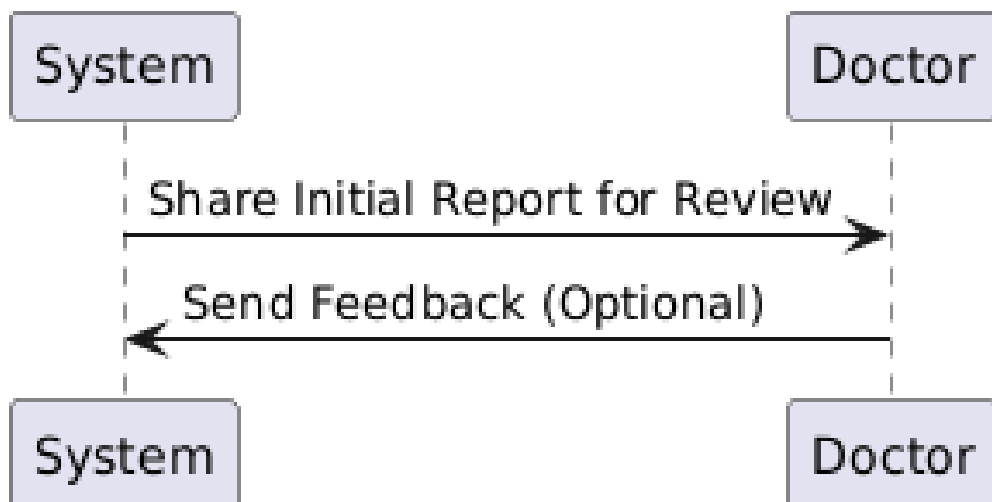


Figure 3.12: UC 3: SD: 3

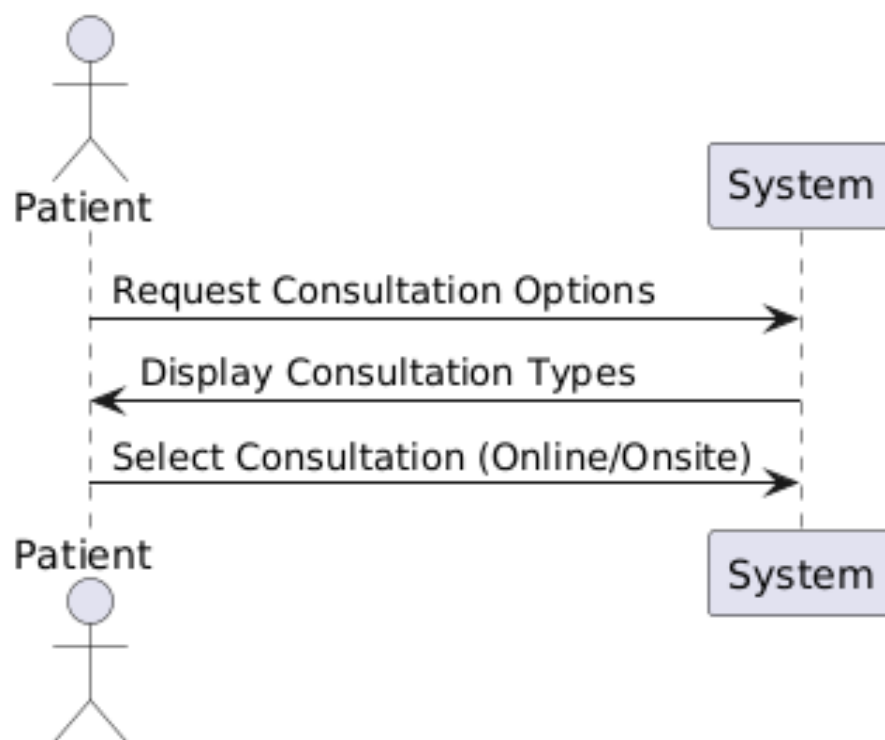


Figure 3.13: UC 4: SD: 4

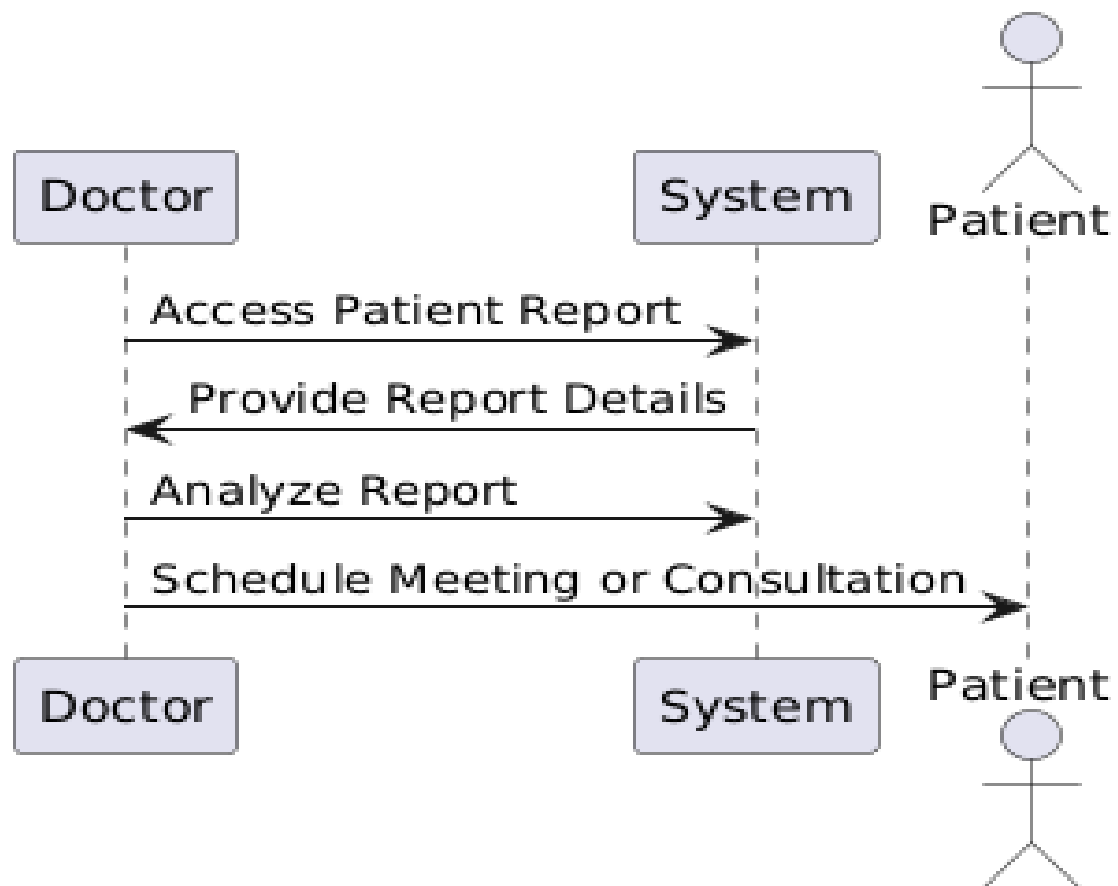


Figure 3.14: UC 5: SD: 5

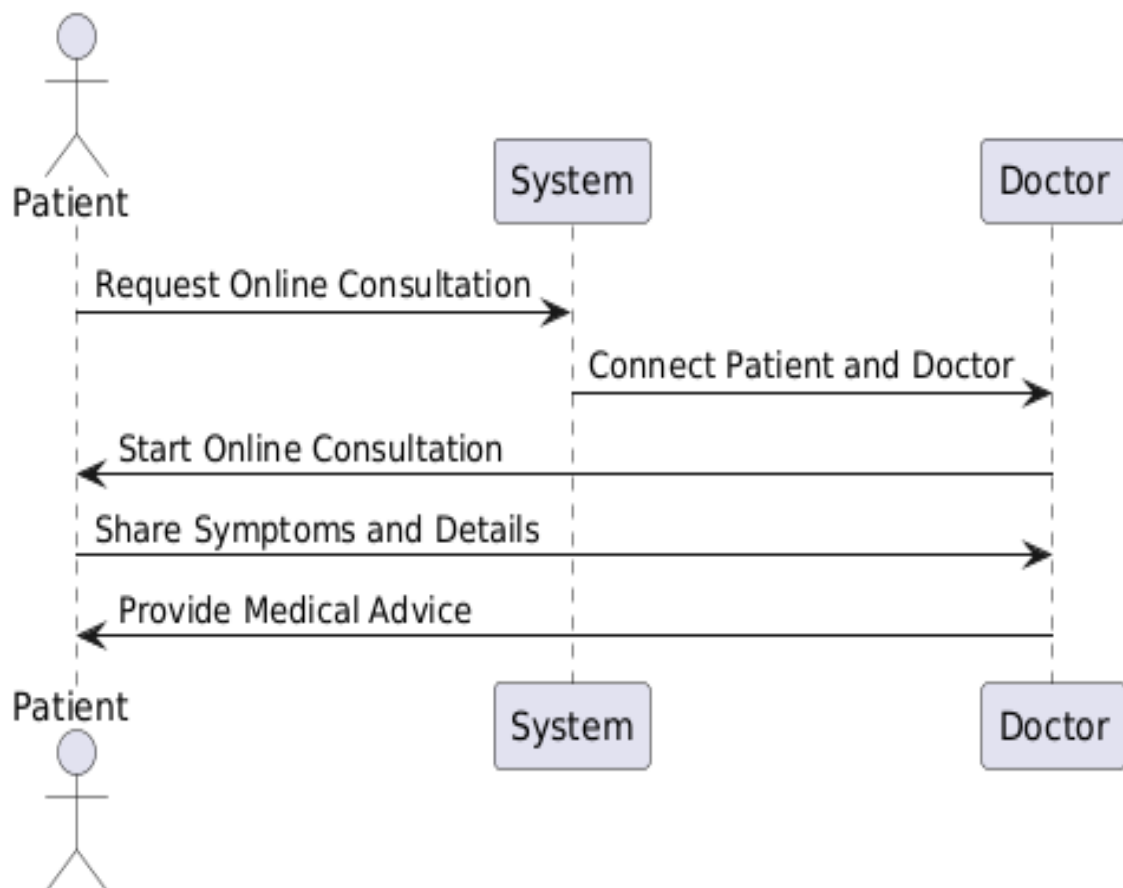


Figure 3.15: UC 6: SD: 6

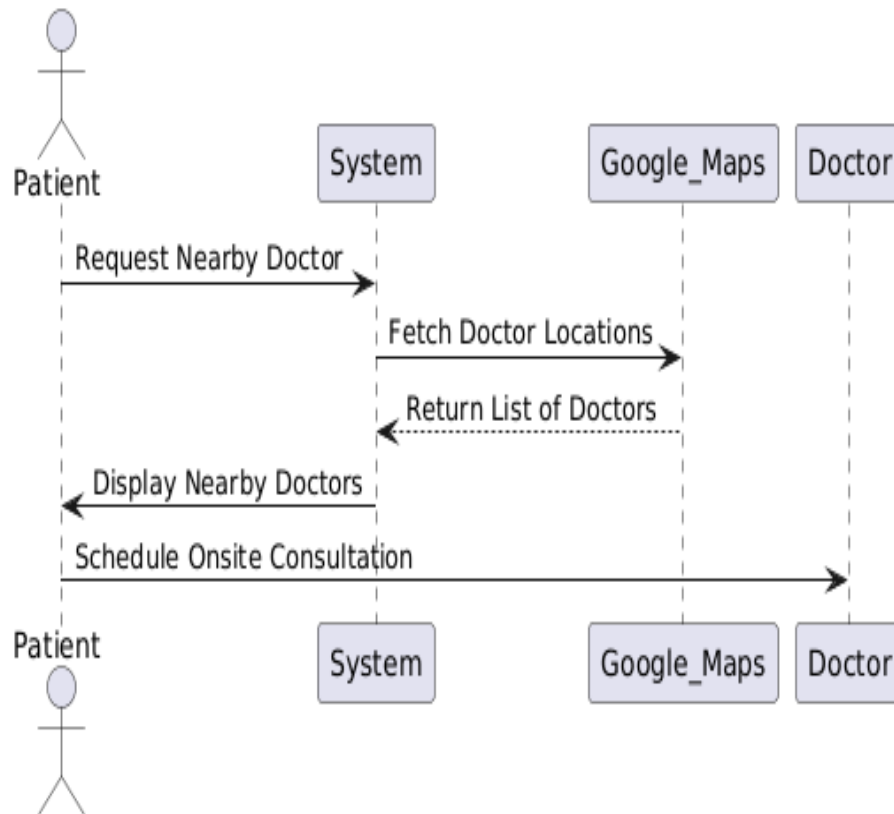


Figure 3.16: UC 7: SD: 7

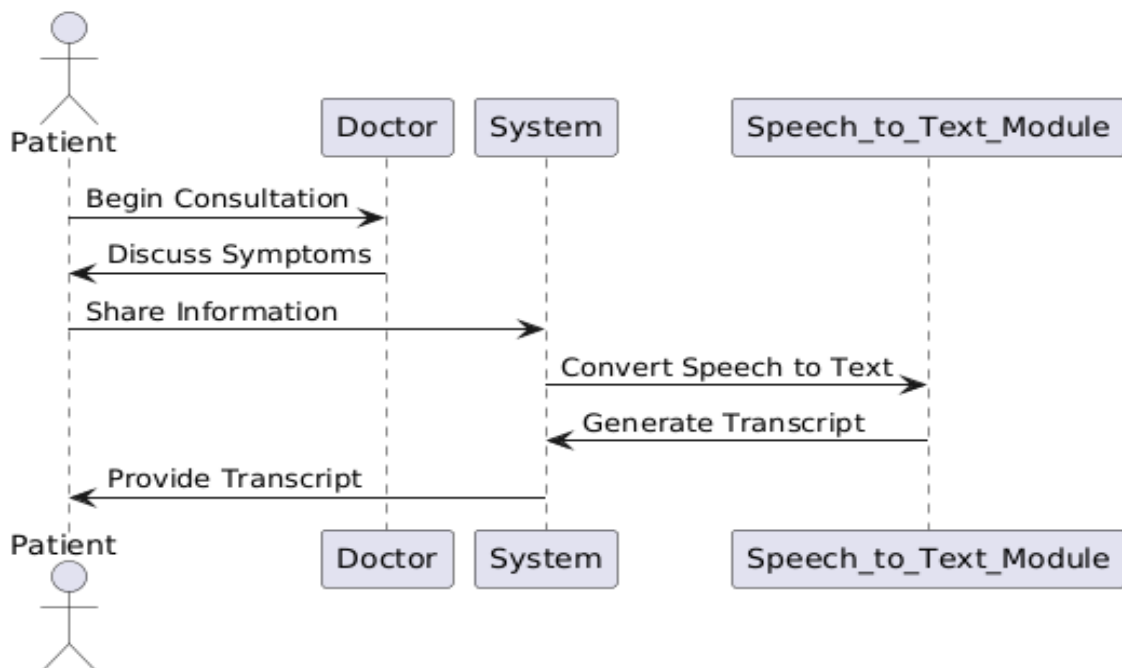


Figure 3.17: UC 8: SD: 8

3.2.4 Class Diagram:

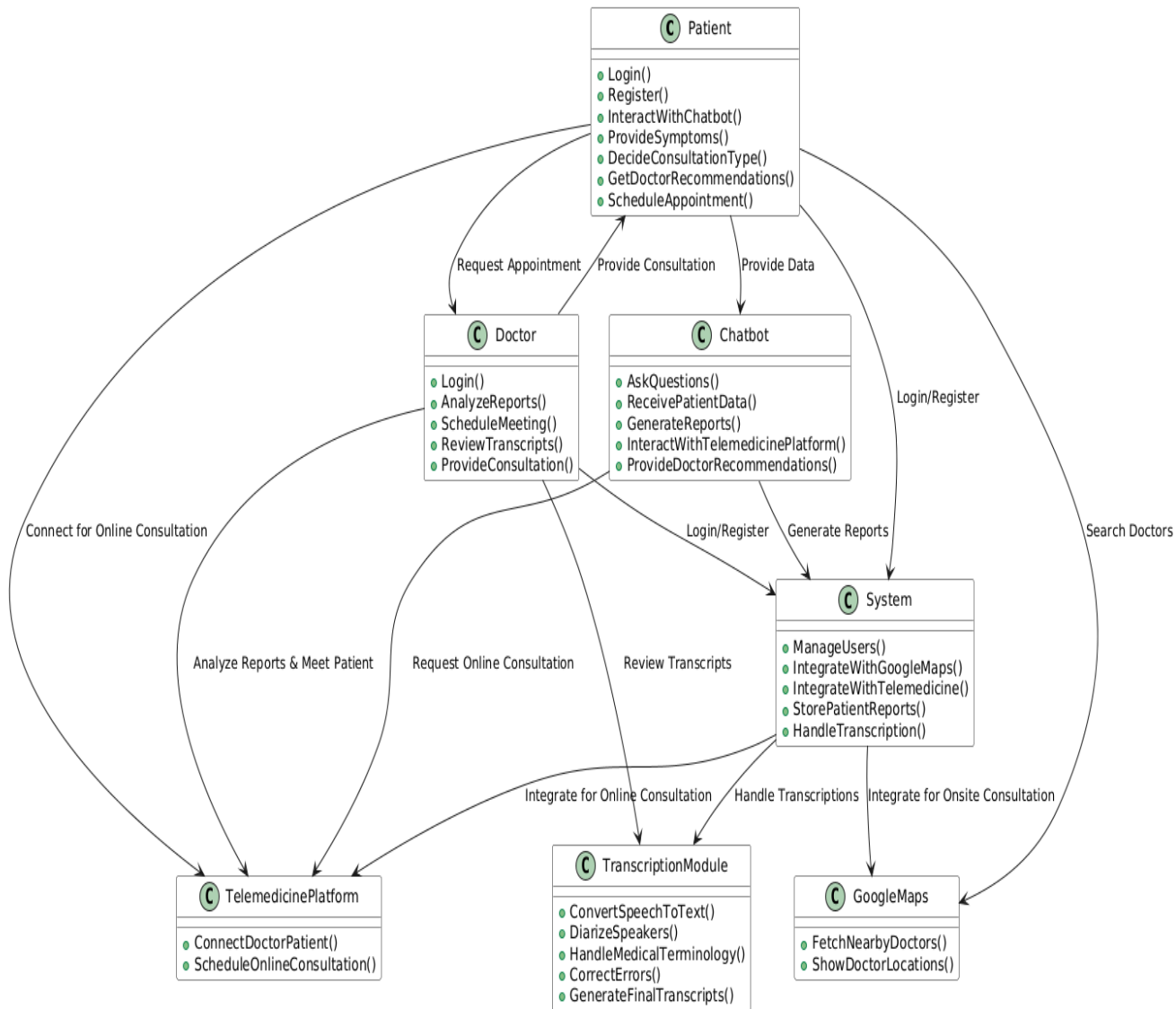


Figure 3.18: Class Diagram

3.3 Data Design

3.3.1 ERD:

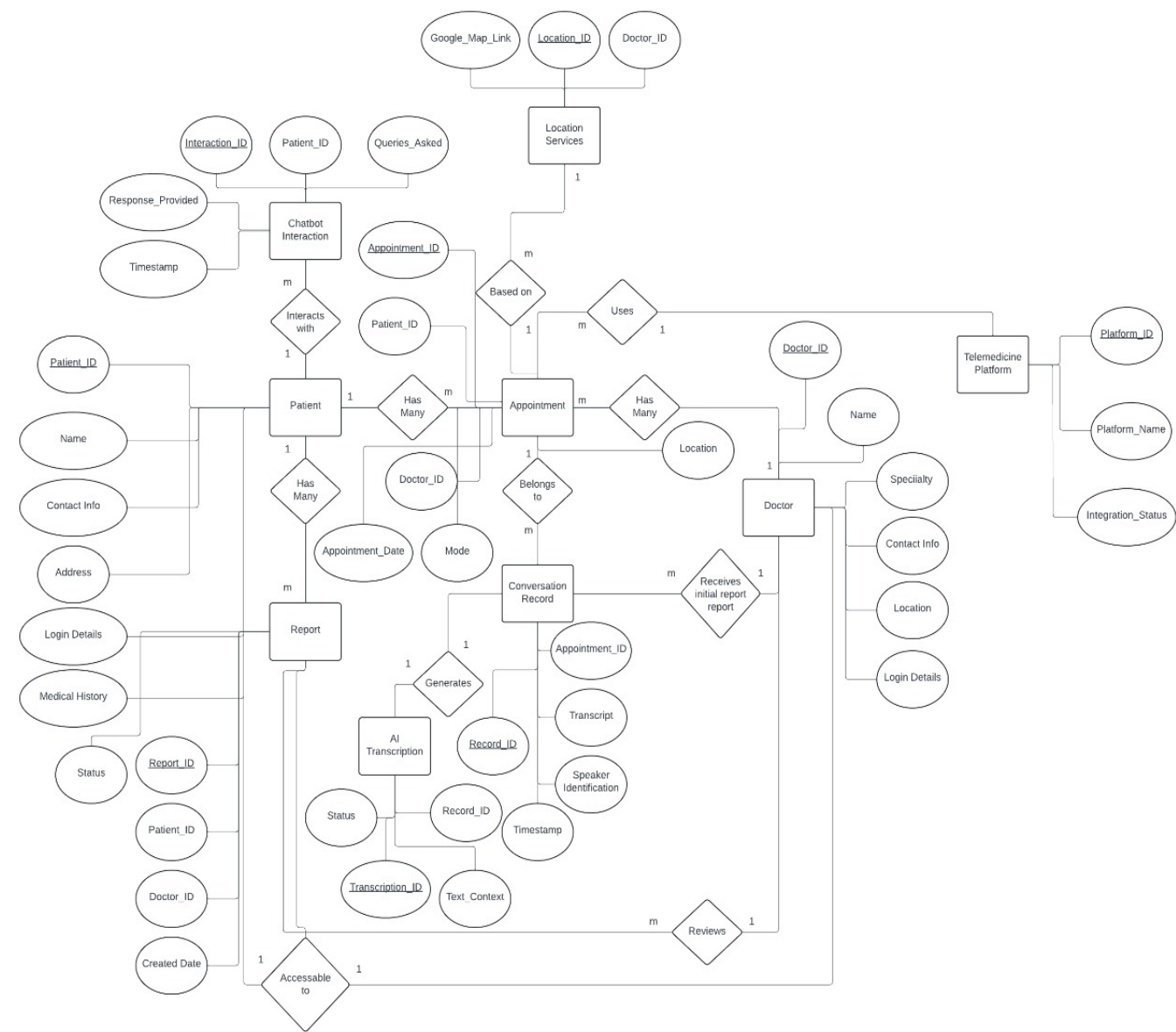


Figure 3.19: ERD

3.3.2 Description:

Entities and Relationships for ERD:

1. Patient:

Attributes:

Patient ID (Primary Key): A unique number of the patient. Name: The patient's name. Contact Information: Contact number, email, etc. Address: Residential address. Login Information: Username and patient password to access a patient portal. Medical History: History of pre-existing diseases, treatments etc.

Relationships:

Has many Appointments: A patient can have multiple appointments scheduled, linked by Appointment ID.

Chatbot Interaction: Any patient can interact with the chatbot for collection of preliminary data.

Has many Reports. Report or multiple reports from a single patient during consultations or generated by the chatbot are managed, and each is linked by a Report ID.

2. Doctor:**Attributes:**

Doctor ID (Primary Key): A unique identifier for each doctor. Name: Name of the doctor.

Speciality: Specialist area of medicine (e.g., cardiologist, neurologist).

Contact Details: Phone number, email, etc.

Location: Work address to visit the site.

Login details: Username and password to the portal of the physician.

Relationships:

Multiple Appointments: A physician has multiple appointments: each appointment is matched to an Appointment ID.

Reviews many Reports: Many Reports are reviewed by a doctor to decide on treatment, and each is keyed by Report ID.

Receives First Report from Chatbot Interaction: The first report produced by the chatbot will then be referred to the relevant doctor for analysis.

3. Appointments:**Attributes:**

Appointment ID: A unique number to that specific appointment.

Patient ID (Foreign key): It refers to the patient connected with the appointment.

Doctor ID (Foreign Key): An appointed doctor to whom a patient is assigned.

Date of Appointment : Appointed date and time.

Mode (Online/Onsite): This sets whether the consultancy process is online or on-site.

Location (if Onsite): Address of the onsite interview.

Relationships:

Related to a Patient and a Doctor: Every contact is related to one doctor and to one patient.

Has associated Conversation Record: The consultation generates at least one conversation records in the process of doctor-patient interaction.

4. Meeting Recording:

Attribute:

Record ID (Primary Key): Unique identifier for every record of conversation.

Appointment ID (Foreign Key): Links the record to the specific appointment.

Transcript (Text): Text transcript of conversation between the doctor and the patient.

Speaker Identification (Doctor/Patient): Identifies who the speaker is for each section of the transcript.

Timestamp: Date and time of conversation.

Relationships:

It Belongs to an Appointment: A given conversation record is related to a specific appointment.

Generates AI Transcription: The AI system will transcribe the record of the conversation for any future reference.

5. AI Transcript:

Attributes:

Transcription ID (Primary Key): This uniquely refers to each transcription.

Record ID (foreign key): Refers to the conversation record.

Text Content: The text content transcribed into the conversation.

Status (Initial/Rectified): It reveals whether the transcription is a fresh copy, or has already been corrected.

Relationships:

Linked to a Conversation Record: Every of the AI transcriptions is linked to a unique conversation record.

Accessible by both Patient and Doctor after completion: Once composed, it is accessible to the doctor and to the patient. Then it is available to the doctor and the patient after it is completed.

6. **Chatbot Interaction:**

Attributes:

Interaction ID (Primary Key): Unique ID for a specific chatbot interaction.

Patient ID Foreign Key-ID: Referencing the patient who interacts with the chatbot.

Questions Asked (Text): The questions asked by the chatbot during the conversation.

Response Patient: Responses elicited by the patient.

Time: Time and date of the encounter.

Relationships:

Belongs to a Patient: Every message belongs to one patient.

Generates an Initial Report for Doctor analysis: The chatbot interaction generates an initial report that gets sent to the doctor.

7. **Report:**

Attributes:

Report ID (Primary Key): Unique number assigned to each report.

Patient ID (Foreign Key): Relates a report to an assigned patient.

Doctor ID Foreign Key: This will associate the report with the relevant doctor.

Created Date: Date the report was generated.

Report Content: The report content, including medical observations, recommendations, prescriptions, etc.

Status (Initial/Updated): Whether the report is an original report or has been updated.

Relationships:

Belongs to Both Patient and Doctor: Every report is linked to the patient as well as to the doctor.

Created post interaction: The report can be created post-interaction with the chatbot or doctor-patient interaction.

Updated during the doctor's analysis process: The doctor was able to update the report during his course of analysis.

8. Telemedicine Platform:

Attributes:

Platform ID: This is a unique number given to each telemedicine platform.

Platform Name: Name of the Telemedicine Platform.

Integration Status: Status of the integration, whether it's complete and on the run.

Relationships:

Facilitates appointment between patient and doctor: An online appointment is possible through the telemedicine platform between patients and doctors.

9. Location Services Features:

Attributes:

Location ID (PK): Unique identifier of the location service.

Doctor ID (Foreign Key): Corresponding to the doctor at the location.

Google Map Link: A Google Maps link to find the doctor's location.

Relationships:

Used for locating nearby Doctors for onsite appointments: The location service was useful for finding nearby Doctors for onsite appointments. This means patients may find doctors near them for onsite consultations.

Relationships Summary:

Patient - Appointment: That is, one patient can have multiple appointments but an appointment has association with only one patient. (Many-to-One) Doctor - Appointment: An appointment may have several doctors where each of the doctors has an appointment. (One-to-Many) Appointment - Conversation Record: One or more conversation records may be attached to every appointment, representing doctor-patient communication. (Many-to-One) Appointment - Telemedicine Platform: An appointment may use a telemedicine platform to conduct a remote consultation. (Many-to-One) Patient-Chatbot Interaction: A patient will have multiple interactions with the Chatbot for gathering initial data. (One-to-Many) Chatbot Interaction - Report: Each chatbot interaction can generate a report that is later shared with the doctor. (One-to-One) Conversation Record - AI Transcription: Each conversation record has a corresponding transcription, rectified and stored for later reference. (One-to-One)

Chapter 4

Appendix

4.1 Multilingual (English and Urdu).

4.2 Integration with Electronic Health Records (EHR) systems.

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